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MARAG-Fin: An Intelligent Multi-agent RAG-LLM Architecture Integrating Financial News Sentiment and Time Series Data for Data-driven Trading Decision-making

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Abstract: Financial markets present a highly dynamic and information-rich environment where effective decision-making requires both accurate forecasting and strong reasoning capabilities. Traditional quantitative models often struggle to capture unstructured knowledge from documents, while large language models (LLMs) excel in reasoning but lack reliable temporal forecasting. To address this gap, this research examines the effectiveness of a multi-agent framework based on Agentic Retrieval-Augmented Generation (RAG) for document-based reasoning, stock index prediction, and investment decision-making. The system combines LLM agents with contextual retrieval and a NeuralProphet-based dynamics agent for price forecasting, creating an adaptive framework suited to market dynamics. Evaluation focuses on three dimensions: reasoning quality, predictive accuracy, and backtested investment strategies on the S&P 500 and NASDAQ-100. Results show that ReAct Prompting outperformed Zero-shot Prompting, achieving an overall reasoning score of 0.92, while GPT-4.1 reached 0.94. The dynamics agent produced accurate forecasts with MAPE of 0.038-0.039. Backtests demonstrated that the S&P 500 strategy achieved a 19.85% return and 28.32% CAGR with stronger risk-adjusted performance, whereas the NASDAQ-100 delivered 57.19% returns with higher volatility. These findings highlight that integrating Agentic RAG, ReAct Prompting, and dynamics agents yields an adaptive and effective multi-agent system for investment decisions.

Keywords: Multi-agent systems, Large language models, Retrieval-augmented generation, Stock index prediction, Investment decision-making, Financial news sentiment, Time series data.

1. Introduction

The increasing complexity and volume of financial data demand more adaptive and efficient analytical methods. In stock markets, where timely and accurate insights are critical for decision-making [1], rapid responsiveness to market dynamics has become a decisive competitive advantage. Global financial participation continues to rise, with data from the Federal Reserve and Gallup (2024) indicating that 62% of U.S. adults now own stocks, up from 55% in 2020, while foreign investors hold about 18% of U.S. equities. This growth, driven by digital trading platforms and low-cost brokerage services, reflects both financial inclusion and the globalization of capital markets. Despite this progress, predicting stock prices and indices remains a complex challenge due to market volatility and the influence of diverse factors such as corporate fundamentals, investor sentiment, macroeconomic indicators, and geopolitical events [2, 3]. Traditional models relying solely on numerical data often fail to capture unstructured textual information from financial news and reports. Early research by Tetlock [2] demonstrated that financial news sentiment impacts market behaviour, but integrating heterogeneous data sources remains difficult, and AI-driven trading systems often suffer from low interpretability and transparency [4].

Machine learning and deep learning approaches have advanced stock trend prediction, yet they struggle to integrate textual and numerical

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information effectively [5]. Recent developments in Large Language Models (LLMs) have enhanced financial text understanding but limitations such as hallucination and reliance on static datasets undermine reliability [5]. Retrieval-Augmented Generation (RAG) addresses these issues by combining internal LLM knowledge with external data sources to improve accuracy and factual grounding [6]. Meanwhile, Multi-Agent Systems (MAS) enable distributed reasoning through collaboration among specialized agents [7- 9]. Integrating RAG and MAS leads to an intelligent architecture known as Agentic RAG, which supports iterative reasoning, adaptive knowledge retrieval, and dynamic task allocation for comprehensive financial analysis [10].

Building on the cognitive systems framework, this study draws parallels between layered human memory and the information organization within trading agents [1]. Simulating multiple memory layers enhances event retrieval, reasoning consistency, and adaptive decision-making in dynamic market conditions [1]. Integrating agents with distinct analytical roles and risk profiles enables adaptive, human-like investment behaviour and improved discovery of market opportunities. Existing Multi-Agent RAG-LLM studies, however, remain limited in data integration, interpretability, and adaptability to evolving market conditions. To address these gaps, this study proposes MARAG-Fin, a collaborative multi-agent RAG framework for U.S. stock trading with a focus on the S&P 500 index. MARAG-Fin integrates financial news sentiment, macroeconomic indicators, deep learning-based time series models, and historical stock data to produce real-time, data-driven investment decisions. The framework leverages LLM-based contextual retrieval for improved predictive accuracy, and Chain-of-Thought (CoT) reasoning is incorporated to enhance transparency and interpretability. The principal contributions of this work include:

1. Development of MARAG-Fin, a Multi-Agent RAG-based framework that integrates financial news sentiment, macroeconomic indicators, and historical stock data for real-time, data-driven trading decisions.
2. Design of four specialized agents (Dynamics, News, Macroeconomic, and Signal) that combine technical, sentiment, and fundamental analyses to emulate human-like reasoning.
3. Implementation of a DPDE-coordinated multi-agent framework with an LLM-based weighted aggregation mechanism, enabling adaptive, transparent, and balanced integration of agent outputs for reliable investment recommendations.

To guide the reader, the remainder of this manuscript is organized as follows. Section 2 reviews related work that provides the theoretical and methodological foundation for this study. Section 3 presents the proposed methodology, detailing the architecture and workflow of the MARAG-Fin framework. Section 4 describes the experimental setup, and Section 5 outlines the evaluation metrics used to assess system performance. Section 6 presents the experimental results, including assessments of reasoning quality, prediction accuracy, investment strategy effectiveness, and a regime-based ablation study across different market conditions. Finally, Section 7 concludes the paper and discusses directions for future research.

2. Related works

Stock market analysis underpins investment and risk management by evaluating economic indicators, financial ratios, and market trends, with studies such as Nagy et al. [11] highlight that understanding the drivers of market performance enhances risk control, opportunity identification, and promoting overall market stability. However, traditional quantitative models struggle to capture sentiment-driven market behaviour, despite evidence that financial news, disclosures, and macroeconomic reports strongly influence investor decisions. BERT-based sentiment models improve prediction when combined with market indicators [12, 13], and the S&P 500 remains sensitive to macroeconomic, technological, and geopolitical factors [11]. LLMs such as GPT, BERT, FinGPT [14], and BloombergGPT [9], along with domain-specific variants like FinBERT [15], expand the capacity to process both structured and unstructured financial data [16], enabling stronger sentiment interpretation for trading and risk management [17, 18]. When paired with numerical forecasting, these models further enhance predictive accuracy [18, 19], though challenges of interpretability, data privacy, and computational cost persist, motivating solutions such as multi-agent systems and RAG frameworks [18, 20].

LLMs leverage transformer architectures to capture financial context more effectively than traditional statistical or shallow learning approaches [9, 14, 16, 18]. Yet, reliance on static training corpora can lead to hallucinations and factual errors. Retrieval-Augmented Generation (RAG) addresses this by grounding LLM outputs in dynamically retrieved external information [21- 23], and has been shown to boost sentiment prediction accuracy by up

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to 48% over standard text-only approaches [24]. While single-agent RAG systems improve reliability, they can lack scalability in fast-changing markets. Multi-agent RAG frameworks counter these limitations by distributing analytical tasks among specialized agents, improving transparency, adaptability, and decision quality under volatile market conditions, as demonstrated by Salve et al. [25].

Time series forecasting remains central to financial prediction, and NeuralProphet offers a robust method by combining Prophet's decomposition with neural network flexibility. It operates on PyTorch, uses stochastic gradient descent, and outperforms Prophet with 55%-92% higher accuracy while preserving interpretability, an essential requirement for financial decision-making [26]. However, NeuralProphet does not natively incorporate real-time sentiment or contextual signals, which justifies integrating it with LLM- and RAG-based systems. Multi-Agent Systems (MAS) further enhance this integration by enabling task specialization across sentiment analysis, technical evaluation, macroeconomic assessment, and predictive modelling [25]. Prior work by Han et al. [7] and Singh et al. [22] shows that multi-agent reasoning improves predictive accuracy beyond single-agent approaches, and incorporating RAG strengthens contextual grounding by retrieving real-time filings, earnings calls, and news [22]. Combined MAS-RAG-LLM architectures thus overcome the constraints of static and opaque models, offering higher accuracy, interpretability, and responsiveness in modern stock market forecasting [7, 22, 25].

3. Methodology

This study presents a multi-agent framework for stock market analysis that integrates RAG with LLMs to improve trading decision accuracy. The system combines financial news, macroeconomic indicators, and historical stock data to produce context-aware insights. Fig. 1 illustrates the overall research workflow and key stages of system development.

3.1 Data acquisition pipeline

3.1.1. Data sources

The primary data source for this study is Yahoo Finance, specifically its Yahoo News feature, which aggregates financial market news from reputable global outlets such as Associated Press, Reuters, Fox News, Al Jazeera, ABC News, USA Today, CNN, and BBC News. With broad coverage and real-time updates, Yahoo News provides rich information on market sentiment, economic developments, and geopolitical events that may influence stock index movements. Its reputable media partners further ensure the credibility and reliability of the data used in the system.

3.1.2. Automated data collection and content structuring

Financial news data are automatically retrieved from Yahoo Finance using Python-based web scraping with the Requests and Selenium libraries. Key elements, including article titles, publication timestamps, sources, and main text, are extracted and organized into structured formats (JSON or CSV) for efficient processing. The cleaned and normalized datasets are stored locally and serve as input to the embedding and vector storage pipeline. Prior to embedding, each news document is segmented into chunks of up to 512 tokens, ensuring compatibility with model processing capacity while preserving contextual coherence.

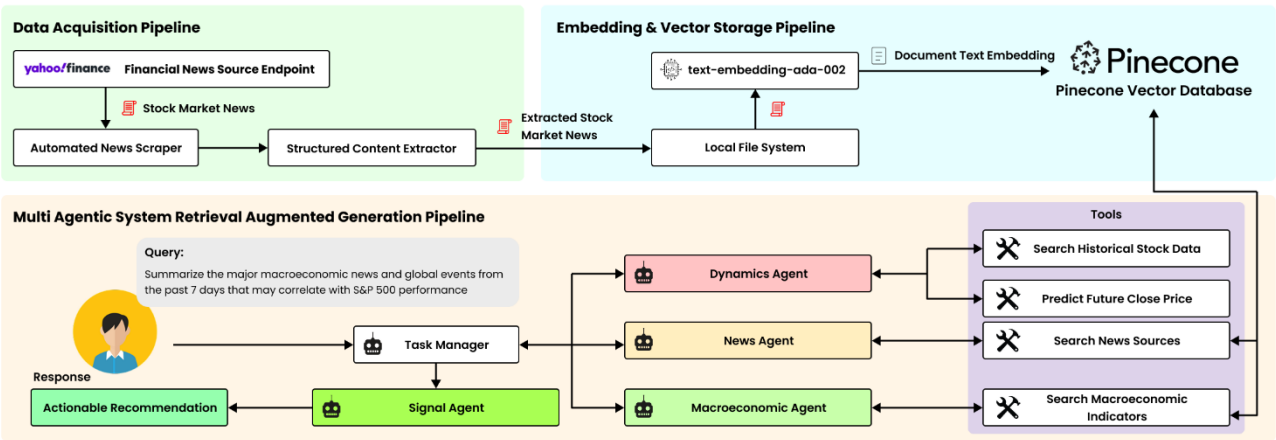

Figure. 1 Architecture of the Proposed MARAG-Fin Multi-Agent RAG-LLM Framework

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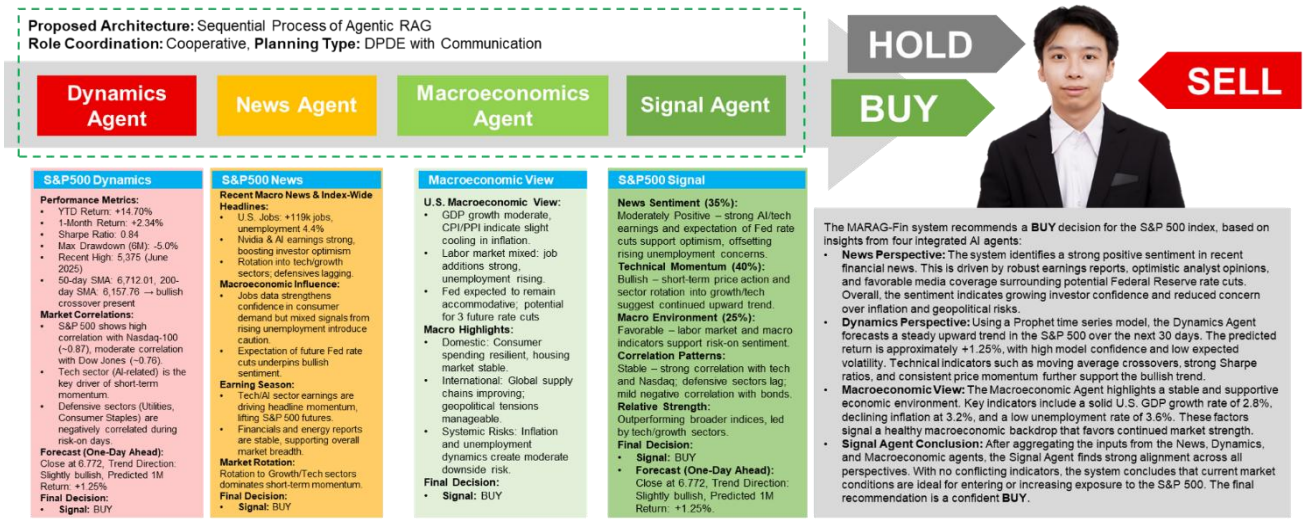

Figure. 2 Analytical Structure and Agent Roles in the MARAG-Fin System (i.e., S&P 500 on Nov 20, 2025).

As highlighted by Bhat et al. [27], chunk sizes between 512 and 1024 tokens are particularly effective for long-context tasks. The entire data acquisition process is automated and continuous, adapting to market dynamics to maintain the relevance and quality of financial information used for investment decision-making.

3.2 Embedding and vector storage pipeline

3.2.1. Document embedding process

Each financial news document from the data acquisition pipeline is embedded using OpenAI's text-embedding-ada-002 model, selected for its strong performance in capturing semantic relationships within text. The model converts each document into a fixed-dimensional vector representation that encodes its contextual meaning, enabling the system to retrieve semantically related information even without direct keyword overlap.

3.2.2. Vector storage architecture with separate indexing

All textual embeddings are stored in Pinecone, a cloud-based platform optimized for large-scale vector storage and retrieval. To enable efficient and domain-specific querying, the system maintains two indexes:

1. news-index: Stores embeddings of financial news from Yahoo Finance, allowing the news agent to retrieve semantically related narratives such as market sentiment and reactions to policy changes.
2. macroeconomic-index: Stores embeddings of global macroeconomic data or information, including indicators such as inflation, interest rates, GDP, and unemployment levels. This index provides the macroeconomic agent with access to broader economic trends that may influence overall market conditions.

Prior to storage in Pinecone, embeddings are briefly saved locally for validation, logging, and backup to ensure data integrity during connectivity issues. The embedding pipeline is automated and modular, supporting both real-time and batch updates. This design improves robustness, scalability, and retrieval precision, enabling each agent to operate within an optimized semantic space for timely, contextually relevant financial and macroeconomic information.

3.3 Multi-agent system retrieval-augmented generation pipeline

The core contribution of this study is MARAG-Fin (Multi-Agent Retrieval-Augmented Generation for Financial Market Analysis), an intelligent system that integrates a multi-agent framework within the RAG paradigm. MARAG-Fin enhances financial market analysis by combining LLM-based reasoning with specialized agents operating in a sequential architecture, enabling coordinated information processing, iterative reasoning, and the generation of adaptive investment insights.

3.3.1. Analytical structure and agent roles in MARAG-fin

The MARAG-Fin framework integrates heterogeneous financial data sources, such as market data, news, and macroeconomic indicators, into a unified system for generating investment recommendations. It employs a modular and

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collaborative architecture in which specialized agents perform distinct analytical tasks and interact to produce comprehensive, interpretable market insights. Based on the Decentralized Planning and Decentralized Execution (DPDE) model (see Section 3.3.2), each agent operates autonomously while maintaining synchronized communication, ensuring flexibility, scalability, and transparency. To achieve a holistic market assessment, MARAG-Fin incorporates four specialized agents, each addressing a unique analytical dimension:

1. Dynamic Agent: Performs quantitative analysis of the S&P 500 index using metrics such as Year-to-Date (YTD) return, Sharpe ratio, maximum drawdown, moving averages, and cross-sector correlations to identify trends, momentum, and volatility. It also employs NeuralProphet for short-term time-series forecasting. Outputs include trading signals, projected returns, and directional trend indicators, forming the quantitative foundation for the system.
2. News Agent: Processes real-time financial news from sources such as Yahoo Finance, Bloomberg, and Reuters. Daily articles are distilled into concise summaries and integrated with prior summaries to construct a progressive narrative, preserving ongoing developments such as policy changes, litigation, or geopolitical events. Through CoT reasoning, the agent evaluates causal relationships and potential impacts on the S&P 500, producing descriptive narratives that capture investor expectations and behavioural dynamics.
3. Macroeconomic Agent: Analyzes domestic and global macroeconomic conditions, including GDP growth, inflation, interest rates, unemployment, and PMI, along with strategic macroeconomic news such as central bank announcements and commodity price dynamics. Scenario-based forecasting anticipates policy shifts and external shocks. Outputs provide assessments of economic stability, systemic risks, and policy directions, offering a medium- to long-term perspective that complements the other agents.
4. Signal Agent: Integrates outputs from the Dynamics, News, and Macroeconomic Agents through a weighted aggregation mechanism, dynamically adjusting the influence of each agent according to market conditions. An LLM-based Weighted Aggregation mechanism is implemented within the Signal Agent, enabling it to evaluate, prioritize, and combine outputs from other agents based on contextual relevance, thereby generating balanced and actionable investment recommendations. It resolves conflicts between divergent signals and produces final outputs comprising buy, hold, or sell recommendations, accompanied by explanatory narratives detailing each agent's contribution to enhance interpretability and credibility.

Algorithm 1 presents the workflow of MARAG-Fin, an intelligent multi-agent system that generates actionable and interpretable investment recommendations.

Algorithm 1: MARAG-Fin Workflow with LLM-Based Weighted Aggregation and Chain-of-Thought Reasoning

Input: financial corpus $\mathcal{F}$, stock user question Q, large language model $\mathcal{M}$, document retriever $\mathcal{T}$, analytical agent set $\mathcal{A} \in \{\mathcal{A}_{\mathrm{D}}, A_{\mathrm{N}}, A_{\mathrm{M}}\}$, signal agent $\mathcal{A}_{\mathrm{S}}$, top similar document k

Output: signal recommendation and concise rationale $\tilde{\mathcal{R}}$

Begin

Initialize LLM $\mathcal{M}$

Initialize retriever $\mathcal{T}$ and vector databases D for financial corpus data $\mathcal{F}$

Deploy agent $\mathcal{A}$

Initialize the information context history $\mathcal{H} \leftarrow \emptyset$

Embed the user query Q and retrieve relevant documents from $\mathcal{F}$ to $\mathcal{T}$

Store retrieved results in $\mathcal{H}$ for all agent's reference

For each analytical agent $\mathcal{A}$ **do**

  Retrieve the top-k most similar documents from the vector database: $D_k \leftarrow \mathcal{T}.\text{similarity_search}(Q, k)$

  Add the retrieved context D_k to the shared information context: $\mathcal{H} \leftarrow \mathcal{H} \cup D_k$

  Perform domain-specific Chain-of-Thought (CoT) reasoning using query Q, shared context $\mathcal{H}$, and language model $\mathcal{M}$: $r_k \leftarrow \mathcal{A}_k.\text{reason}(Q, \mathcal{H}, \mathcal{M})$

  Generate domain-specific analytical output: $\mathcal{O}_k \leftarrow \mathcal{A}_k.\text{analyze}(Q, r_k, \mathcal{F})$

  Append both the analytical output and the reasoning trace to the shared context: $\mathcal{H} \leftarrow \mathcal{H} \cup \{(\mathcal{O}_k, r_k)\}$

End For

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13: Agents $\mathcal{A}$ exchange and review CoT summaries for coherence. Refined outputs $\mathcal{O}_k \leftarrow \mathcal{A}_k.\text{refine}(\mathcal{O}_k, \mathcal{H})$
14: Signal Agent $\mathcal{A}_S$ performs weighted aggregation: $\tilde{\mathcal{R}} \leftarrow \mathcal{A}_S$ .aggregate $(\{\mathcal{O}'_k, r_k\}, w_k)$ where w_k are confidence weights derived from CoT coherence
15: Generate the signal recommendation $\tilde{\mathcal{R}}$ using $\mathcal{M}$ given Q and $\mathcal{H}$ along with a concise natural-language rationale derived from the CoT summaries
16: **Return** $\tilde{\mathcal{R}}$
End

It integrates multiple financial data sources and employs an LLM-based Weighted Aggregation mechanism, where a single Large Language Model dynamically evaluates the relevance and reliability of each agent's output to determine its influence on the final decision. Unlike traditional meta-agent methods with fixed weights or rule-based integration, this adaptive approach resolves conflicting signals in real time, producing coherent, context-aware investment guidance along with explanatory summaries of each agent's contribution. To enhance transparency, Section 4 outlines the mathematical formulation of the forecasting component, implemented using the NeuralProphet model as part of the MARAG-Fin framework.

3.3.2. Coordination and execution mechanism

MARAG-Fin employs a cooperative coordination model in which agents perform specialized analytical tasks while collaborating to achieve shared objectives. The system follows a Decentralized Planning and Decentralized Execution (DPDE) approach with communication, in which each agent plans independently, executes tasks in parallel or sequentially as required, and exchanges analytical outputs to maintain overall system coherence. This mechanism provides three key advantages: (1) adaptivity, as agents autonomously adjust to variations in input data and market dynamics; (2) transparency, through traceable reasoning and decision processes that facilitate monitoring and evaluation; and (3) modularity, enabling independent development and refinement without disrupting the system. Together, these features establish MARAG-Fin as a robust, scalable, and maintainable framework for comprehensive financial market analysis.

3.3.3. System output: comprehensive and consistent investment recommendations

Upon completion of the multi-agent analysis, MARAG-Fin produces integrated outputs consisting of comprehensive analyses and final investment recommendations. These recommendations are classified into three actionable categories:

1. Buy: Indicating strong signals of anticipated market appreciation based on convergent positive indicators.
2. Hold: Reflecting neutral conditions or uncertainty regarding market direction, with insufficient justification for action.
3. Sell: Signalling potential market corrections or downward pressures, inferred from negative outputs of technical, sentiment, and macroeconomic analyses.

By combining quantitative, sentiment-driven, and fundamental dimensions, MARAG-Fin generates recommendations that are both analytically rigorous and operationally practical for informed investment decision-making.

4. Experimental setup

The experimental configuration in this study is designed to evaluate the performance of an intelligent multi-agent system for market analysis and trading signal forecasting. The system integrates LLMs, embedding models, vector databases, and a structured prompt engineering strategy. The experiments aim to examine how these components interact within a collaborative agent architecture to enhance reasoning capability, adaptivity, and predictive accuracy. Importantly, the experimental design follows a layered and progressive evaluation framework, where the best-performing results from each stage are systematically used as the foundation for the next. This iterative structure ensures continuous optimization of the system, both in predictive performance and in practical applicability, through cumulative refinement of the most effective configurations. The experimental workflow comprises four main stages: (1) evaluating prompt engineering strategies, (2) comparing LLM performance, (3) assessing prediction horizons, and (4) testing portfolio performance. Each stage builds upon the results of the preceding one, forming a coherent and structured evaluation process.

4.1 Prompt engineering strategy

A systematic prompt engineering framework was developed to support coherent, role-based, and

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context-aware reasoning among the agents. Two main prompting strategies were applied:

a. Zero-shot Prompting: Serves as the baseline, generating responses solely from given instructions without prior examples to assess the model's inherent reasoning capability.
b. ReAct Prompting: The advanced strategy integrates Reasoning and Acting (ReAct) with Chain of Thought (CoT) reasoning to foster more complex, transparent, and adaptive reasoning behaviours. ReAct prompting allows the model to articulate intermediate reasoning and corresponding action steps, while CoT guides step-by-step logical processing akin to human reasoning patterns.

Both strategies incorporate three complementary prompt patterns: role prompting, instruction prompting, and template prompting, which maintain consistency and reliability. Role prompting defines each agent's analytical scope, instruction prompting specifies task objectives, and template prompting ensures structural uniformity, including context and expected outputs. Each agent (Dynamics, News, Macroeconomic, and Signal) uses customized prompts designed through structured prompt chaining, enabling reasoning outputs to flow sequentially between agents. This structured design enhances inter-agent coordination, contextual consistency, and the overall coherence of the multi-agent reasoning process.

4.2 Large language models and comparative analysis

Two models from the GPT-4.1 family, GPT-4.1-nano and GPT-4.1, were used to assess the impact of model scale on reasoning quality and predictive performance. GPT-4.1-nano serves as the baseline model, offering computational efficiency and stable reasoning despite its smaller capacity, while GPT-4.1, with its larger context window and enhanced reasoning capabilities, functions as the comparative model. This analysis examines how differences in model size and capacity affect reasoning depth, response consistency, and predictive accuracy.

4.3 Prediction models and comparative evaluation

This stage evaluates the system's forecasting capability and its applicability in real investment contexts. Using the best-performing configurations from earlier stages, the predictive framework is tested for accuracy and robustness under real market conditions. Forecasting performance is assessed across two horizons: one-day-ahead (short-term) and five-day-ahead (medium-term) predictions. The predictive engine employed for this purpose is NeuralProphet, a neural network-based time-series model capable of capturing complex and nonlinear temporal dynamics in financial data. The dataset covers the period from 2015 to 2025, with approximately 90% (2015-2024) used for training and 10% (2025) allocated for testing.

In the final stage, the focus shifts to portfolio performance evaluation, which examines the real-world effectiveness of the predictive signals generated in the forecasting phase. The trading signals are applied to construct investment portfolios based on two major U.S. stock indices: the S&P 500, serving as the baseline, and the NASDAQ-100, serving as the comparative benchmark. Backtesting simulations are performed to replicate real trading scenarios and evaluate the profitability, risk-adjusted returns, and overall performance consistency of the model's strategies. Through this final evaluation, the study validates the system's capacity to generate actionable trading insights and support effective investment decision-making in dynamic and uncertain financial markets.

4.3.1. Time-series forecasting model formulation

The forecasting model used in this study integrates traditional additive time-series decomposition with the flexibility of neural networks. In this approach, the time series is represented as the sum of key components such as trend, seasonality, autoregressive terms, external factors, and event effects, with a neural mapping function added to capture nonlinear patterns in financial data. The general formulation of the forecasting framework is expressed as follows:

$$\hat{y}_t = T(t) + S(t) + E(t) + F(t) + A(t) + L(t) + \epsilon_t \quad (1)$$

In this model, $\hat{y}(t)$ represents the predicted value at time t, built from several interpretable components. The trend, $T(t)$, reflects the overall direction of the series over time, while $S(t)$ captures repeating seasonal patterns. Sudden spikes or drops caused by events or holidays are modeled by $E(t)$. The model also considers external influences: $F(t)$ incorporates future-known factors, and $L(t)$ uses past values of external variables. Short-term dependencies in the series are captured by $A(t)$, and any remaining unexplained variation is captured by the noise term ϵ_t. The detailed component definitions and computational procedures follow the standard

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formulations established in the NeuralProphet framework [26] and are therefore not restated here.

5. Evaluation metrics

5.1 Evaluation of information quality and textual justification

To ensure the credibility and factual soundness of the system's generated responses, the Agentic Retrieval-Augmented Generation (RAG) component was evaluated in terms of information quality and textual justification. The evaluation employed the Retrieval-Augmented Generation Assessment Score (RAGAS) framework [28], a standardized methodology specifically designed for assessing the performance of RAG-based systems. RAGAS provides a systematic means of evaluating how effectively a model integrates retrieved knowledge into generated responses, emphasizing both retrieval accuracy and generation reliability. The assessment focuses on three principal dimensions: Faithfulness, Answer Relevancy, and Context Precision, which together offer a comprehensive understanding of the system's information quality.

1. Faithfulness measures how accurately the system's responses align with the retrieved source material, reflecting its ability to avoid hallucinations or unsupported claims. In knowledge-intensive domains such as finance, high faithfulness ensures that generated content is based on verified information and supports transparent decision-making.
2. Answer Relevancy evaluates the semantic and contextual alignment between generated responses and user queries. It reflects how effectively the model interprets intent and produces directly relevant, contextually appropriate answers, demonstrating a strong level of query comprehension.
3. Context Precision quantifies the proportion of retrieved information that is genuinely relevant to a query, assessing both the efficiency and quality of the retrieval process. High precision reduces informational noise and improves the coherence and focus of generated explanations.

Collectively, these metrics provide a comprehensive assessment of the system's ability to generate accurate, relevant, and well-justified responses, indicating how effectively the RAG component balances retrieval quality with generative reasoning. In this study, the detailed formulations and computational procedures follow the standard definitions established in the RAGAS framework [28] and are therefore not restated here. The overall performance of the Agentic RAG component was calculated by averaging the normalized scores across all metrics, producing a composite measure that reflects the system's capacity to deliver faithful, relevant, and contextually precise outputs, ensuring reliability and interpretability in financial decision-making contexts.

5.2 Error rate evaluation of model prediction accuracy

The second evaluation focuses on assessing the prediction accuracy of the system's Dynamics Agent, which is responsible for forecasting stock index prices and other relevant technical indicators. To ensure a robust and quantitative evaluation, this study utilizes three well-established error metrics in time series forecasting: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE), as referenced in [29].

1. Mean Absolute Error (MAE) measures the average absolute difference between predicted and actual values, providing an interpretable indication of overall forecasting accuracy in the same unit as the data.
2. Root Mean Squared Error (RMSE) calculates the square root of the mean squared differences, penalizing larger errors more heavily and highlighting significant deviations that may occur in volatile markets.
3. Mean Absolute Percentage Error (MAPE) expresses the average percentage error, enabling scale-independent comparison across multiple indices or assets with varying price levels.

This evaluation provides a comprehensive analysis of the model's prediction performance, ensuring a rigorous assessment of its forecasting precision and reliability. The results demonstrate the model's effectiveness as a dependable decision-support component for financial forecasting and investment strategy formulation.

5.3 Financial investment performance evaluation

The third evaluation phase assesses the effectiveness, consistency, and risk-adjusted performance of investment decisions generated by the MARAG-Fin system. These outcomes reflect the combined decision-making of the four intelligent agents within the framework.

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Table 1. Financial Investment Performance Metrics and Descriptions

Metrics	Description
Total Return (TR)	Represents the overall percentage change in portfolio value over the evaluation period, including realized and unrealized gains or losses. It indicates the total profitability achieved through the system's investment decisions.
Win Rate (WR)	The proportion of profitable trades relative to total trades executed. It reflects the accuracy and consistency of the system's trading signals, although it should be considered alongside profit-based metrics.
Sharpe Ratio	Measures excess portfolio return (above the risk-free rate) per unit of total volatility. A higher Sharpe ratio indicates more efficient risk-adjusted performance.
Sortino Ratio	A variation of the Sharpe ratio that considers only downside risk, focusing on negative return deviations. It provides a more accurate measure of performance stability during adverse market conditions.
Annual Volatility (Vol)	Indicates the standard deviation of returns on an annual basis, representing the level of fluctuation in portfolio performance. Lower volatility suggests more stable returns, while higher values imply greater risk exposure.
Maximum Drawdown (MDD)	Measures the largest drop in portfolio value from a peak to a trough before recovery. It highlights the system's worst-case loss scenario and resilience to market downturns.
Daily Maximum Drawdown (MDDd)	Normalizes MDD by the number of days in the decline period, showing the average daily rate of loss. Lower values indicate slower and more controlled drawdowns.
Compound Annual Growth Rate (CAGR)	Reflects the mean annual growth rate of the portfolio over the investment period, accounting for compounding effects. It provides a realistic measure of long-term growth performance.
Profit Factor	The ratio of total profits to total losses from all trades. A value above 1 indicates overall profitability, while a value below 1 signals an unprofitable trading strategy.

Evaluation was conducted using standardized financial metrics commonly employed in quantitative finance and portfolio management, summarized in Table 1, which capture profitability, consistency, risk exposure, and capital efficiency.

Collectively, these metrics form a framework for evaluating the investment soundness of the MARAG-Fin system. They assess profitability, growth sustainability, risk-adjusted performance, stability under market fluctuations, and operational efficiency, enabling a holistic view of the system's potential for sustainable and resilient financial performance.

6. Results and discussion

6.1 Evaluation results of agentic RAG

The evaluation assessed document-based reasoning in the Agentic RAG framework, which combines LLM-driven agents with context-aware retrieval. Two factors were tested: prompting strategies and LLM model variations. Each agent completed 50 trials, and outputs were scored using RAGAS framework. The assessment focused on the News and Macroeconomic Agents, with results shown in Tables 2 and 3.

The evaluation results in Table 2 compare the effectiveness of two prompting strategies, Zero-shot Prompting and ReAct Prompting, implemented with the GPT-4.1-nano model.

Table 2. Performance Evaluation of Agentic RAG Based on Prompt Engineering Methods using GPT-4.1-nano

Prompt Engineering	Faithfulness	Response Relevancy	Context Precision
Zero-shot Prompting	0.55	0.90	1.00
ReAct Prompting	0.86	0.90	1.00

Table 3. Evaluation Results of Agentic RAG Comparing the GPT-4.1-nano and GPT-4.1 LLMs

LLMs	Faithfulness	Response Relevancy	Context Precision
GPT-4.1-nano	0.86	0.90	1.00
GPT-4.1	0.91	0.90	1.00

A notable difference is observed in answer faithfulness, where ReAct Prompting achieved a score of 0.86, significantly outperforming Zero-shot Prompting at 0.55. This result highlights the advantage of ReAct's structured reasoning process, which integrates explicit reasoning steps with targeted information retrieval from the vector database, ensuring that responses remain grounded in the source documents. In contrast, Zero-shot Prompting relies primarily on the model's internal knowledge without such reasoning mechanisms,

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increasing the likelihood of unsupported or partially factual outputs. Both techniques, however, produced comparable scores in response relevance (0.90) and achieved perfect results in context precision (1.00), indicating that both could identify and utilize the most relevant reference documents. Nonetheless, the superior aggregate score of 0.92 for ReAct Prompting, compared to 0.82 for Zero-shot Prompting, confirms its stronger consistency in balancing factual accuracy, contextual alignment, and relevance.

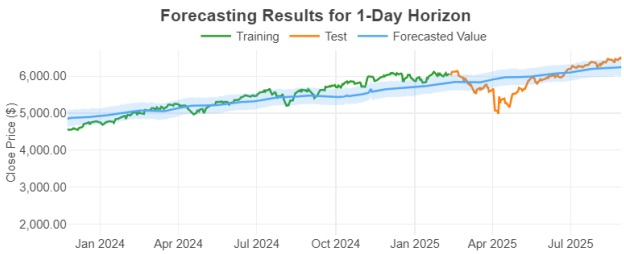
Subsequently, Table 3 presents a comparison between two LLMs, GPT-4.1 and GPT-4.1-nano. The results show that GPT-4.1 consistently outperformed the smaller variant across most dimensions. In terms of faithfulness, GPT-4.1 achieved 0.91, slightly higher than GPT-4.1-nano's 0.86, reflecting its enhanced ability to integrate retrieved evidence due to its larger parameter capacity. Both models performed equally well in response relevance (0.90) and context precision (1.00), demonstrating their reliability in identifying appropriate reference contexts. The aggregate scores, 0.94 for GPT-4.1 and 0.92 for GPT-4.1-nano, further indicate GPT-4.1's superior consistency in generating evidence-based reasoning. Overall, the combination of ReAct Prompting with GPT-4.1 emerged as the most effective configuration within the Agentic RAG architecture. This setup ensures high faithfulness and precise contextual reasoning, making it the most reliable choice for document-based decision support. Meanwhile, the GPT-4.1-nano and Zero-shot Prompting approaches remain viable for resource-constrained applications, though with minor compromises in factual integrity.

6.2 Evaluation results of model prediction accuracy

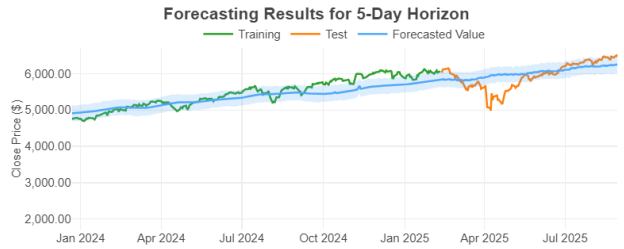
The prediction performance evaluation assessed the time series forecasting capabilities of the Dynamics Agent. The NeuralProphet model was employed to capture historical stock index patterns and generate 1-day and 5-day forecasts. Forecasting accuracy was evaluated by comparing predicted and actual values over daily and weekly intervals using the performance metrics described in Section 5.2. The results of this evaluation are summarized in Table 4.

Table 4. Prediction Performance by Horizon

Prediction Horizons	MAE	RMSE	MAPE
1 Day	219.45	289.72	0.038
5 Days	219.72	290.89	0.039



(a)



(b)

Figure. 3 S&P 500 Index Price Forecasting: (a) One-Day Horizon and (b) Five-Day Horizon

The evaluation of prediction accuracy for the NeuralProphet model implemented within the dynamics agent demonstrates a satisfactory performance in projecting stock index price movements. The evaluation was conducted by comparing actual values with forecasted outputs across two prediction horizons, namely one day (1-day) and five days (5-day). Based on the test results, for the 1-day prediction horizon, the model achieved a Mean Absolute Error (MAE) of 219.45, a Root Mean Squared Error (RMSE) of 289.72, and a Mean Absolute Percentage Error (MAPE) of 0.038, or approximately 3.8%. These values indicate that the average deviation between the predicted results and the actual data is relatively low, suggesting that the model possesses strong accuracy in performing very short-term forecasts. For the 5-day prediction horizon, the model obtained an MAE of 219.72, an RMSE of 290.89, and a MAPE of 0.039, or about 3.9%. These results show only a slight increase in error compared to the 1-day horizon, indicating that the model maintains stable performance over longer prediction periods. The forecasting results are illustrated in Fig. 3.

Fig. 3 (a) illustrates the forecasting performance of the NeuralProphet model with a 1-day prediction horizon. The blue line represents the historical training data, the green line denotes the actual test data, and the orange line shows the model's predicted values, while the light blue shaded area indicates the confidence interval. The visualization demonstrates that the model successfully captures both the overall upward trend and the short-term fluctuations of the

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stock price. The predicted values closely align with the actual data, showing only minor deviations during brief periods of volatility. This strong alignment is further confirmed by the quantitative results, with a MAPE value of 3.8%, indicating a low prediction error. Such performance suggests that the NeuralProphet model is capable of providing accurate and reliable short-term forecasts, which are essential for market participants in making daily trading and investment decisions.

Fig. 3 (b) presents the forecasting results with a 5-day horizon. The general pattern of the predictions remains consistent with the actual stock price movements, demonstrating the model's ability to sustain accuracy over a medium-term period. However, as the prediction horizon extends, slightly larger deviations can be observed, particularly during sharp price fluctuations. This is reflected in the modest increase of MAPE to 3.9%, indicating a minor decline in precision but maintaining an acceptable error level. Overall, the NeuralProphet model exhibits strong capability in capturing long-term trends and seasonal behaviors while showing limited responsiveness to abrupt market volatility. The inclusion of reasonably wide confidence intervals helps account for this uncertainty. With MAPE values consistently below 5% across both horizons, the model proves stable and dependable for short- to medium-term forecasting, supporting its practical use in stock market analysis, investment strategy formulation, and portfolio management.

6.3 Evaluation results of investment strategy effectiveness

The investment strategy evaluation assessed the ability of portfolios derived from the Signal Agent's recommendations to balance return and risk. As the system's final component, the Signal Agent integrates insights from the Dynamics, News, and Macroeconomic Agents via an LLM-based weighted aggregation mechanism, dynamically evaluating and combining agent outputs to produce context-aware and balanced investment recommendations. Evaluation was conducted through backtesting on historical market data, replicating real trading conditions to examine profitability, robustness, and resilience under market volatility, serving as a key validation of the Agentic RAG framework. The effectiveness of the investment strategy was assessed using the financial performance metrics reported in Table 1. The number of trades executed varied across indices, with 350 trades for the S&P 500 and 600 trades for the NASDAQ-100, which may influence the reliability of proportion-based metrics such as win rate. All evaluations employed the best-performing configuration: ReAct Prompting with GPT-4.1 under a one-day trading horizon.

Table 5 summarizes the evaluation results of investment strategy effectiveness for the 1-day prediction horizon across two major indices, the S&P 500 and NASDAQ-100. Based on the win rate, the S&P 500 strategy achieved 57.14% profitable trades (200 out of 350), slightly outperforming the NASDAQ-100's 50% (300 out of 600). However, due to the smaller number of trades in the S&P 500, this metric should be interpreted with caution, as statistical variability may be higher. In terms of total return, the NASDAQ-100 recorded a stronger gain of 57.19%, compared to 19.85% for the S&P 500, indicating that the NASDAQ-100 strategy captured more aggressive growth opportunities. Yet, when adjusted for risk, the S&P 500 strategy demonstrated superior efficiency, with a Sharpe ratio of 1.65 and a Sortino ratio of 3.66, exceeding the NASDAQ-100's 1.17 and 2.57, respectively. This indicates that the S&P 500 strategy achieved higher returns per unit of total and downside risk.

The risk profile analysis further highlights the S&P 500's resilience. Although its annualized volatility (26.14) was slightly higher than that of the NASDAQ-100 (21.43), it experienced a substantially lower Maximum Drawdown (6.88%) compared to 15.02%, reflecting better loss control. Similarly, the Daily Maximum Drawdown (MDDd) lasted only 15 days for the S&P 500 versus 44 days for the NASDAQ-100, suggesting quicker recovery from drawdowns. The Compound Annual Growth Rate (CAGR) and profit factor also reinforced this advantage, with the S&P 500 achieving 28.32% and 7.29, respectively, far exceeding the NASDAQ-100's 3.28% and 3.46.

Table 5. Evaluation Results of Investment Strategy Effectiveness Based on the S&P 500 and NASDAQ-100 Indices with a One-Day Prediction Horizon

U.S. Indices	Win Rate	Total Return	Sharpe Ratio	Sortino Ratio	Volatility	MDD	MDDd	CAGR	Profit Factor
S&P 500	57.14	19.85	1.65	3.66	26.14	6.88	15	28.32	7.29
NASDAQ-100	50.00	57.19	1.17	2.57	21.43	15.02	44	3.28	3.46

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Collectively, these results demonstrate that while the NASDAQ-100 strategy exhibited stronger raw returns, the S&P 500 strategy offered a more stable and risk-efficient performance, making it more reliable for consistent profit generation and long-term portfolio sustainability.

6.4 Ablation study across market regimes

To assess the robustness of the proposed MARAG-Fin architecture under varying market conditions, a regime-based ablation study was conducted using a fixed configuration across all experiments, consisting of ReAct prompting with Chain-of-Thought reasoning, GPT-4.1, and a one-day prediction horizon. The evaluation was organized into three distinct market regimes: (i) a bull regime spanning January 2023 to July 2024, (ii) a crash regime in March to April 2025, and (iii) a recovery regime from May to October 2025. These regimes were selected to capture structurally different patterns in financial behaviour. For each regime, the study examined both predictive accuracy and trading effectiveness using a consistent set of performance metrics, including cumulative return (CR), Sharpe ratio (SR), and maximum drawdown (MDD). This experimental design facilitates a clear and systematic comparison of how MARAG-Fin responds to variations in volatility, market sentiment, and liquidity across the full market cycle.

6.4.1. Ablation on bull market regime

In the bull market regime, characterized by strong upward trends and high investor confidence, MARAG-Fin successfully captured market momentum by combining structured data (e.g., economic indicators) and unstructured data (e.g., news and social media) to guide long-biased strategies. During this period, bull markets were mainly supported by strong performance in AI and tech stocks, easing inflation, expectations of looser monetary policy, and economic data pointing to a soft landing. MARAG-Fin achieved a prediction accuracy of 60%, a cumulative return of +22.2%, a Sharpe ratio of 1.31, and a maximum drawdown of −6.43%. The ReAct component helped the model reason about macroeconomic trends, while Chain-of-Thought (CoT) improved alignment of daily predictions with medium-term momentum. Removing either component reduced cumulative returns by about 7-10%, showing that the model's reasoning steps were crucial for capturing gains in sentiment-driven bull markets.

6.4.2. Ablation on crash market regime

During the crash regime, marked by sharp declines, high volatility, and negative investor sentiment, MARAG-Fin was tested for its ability to manage downside risk.

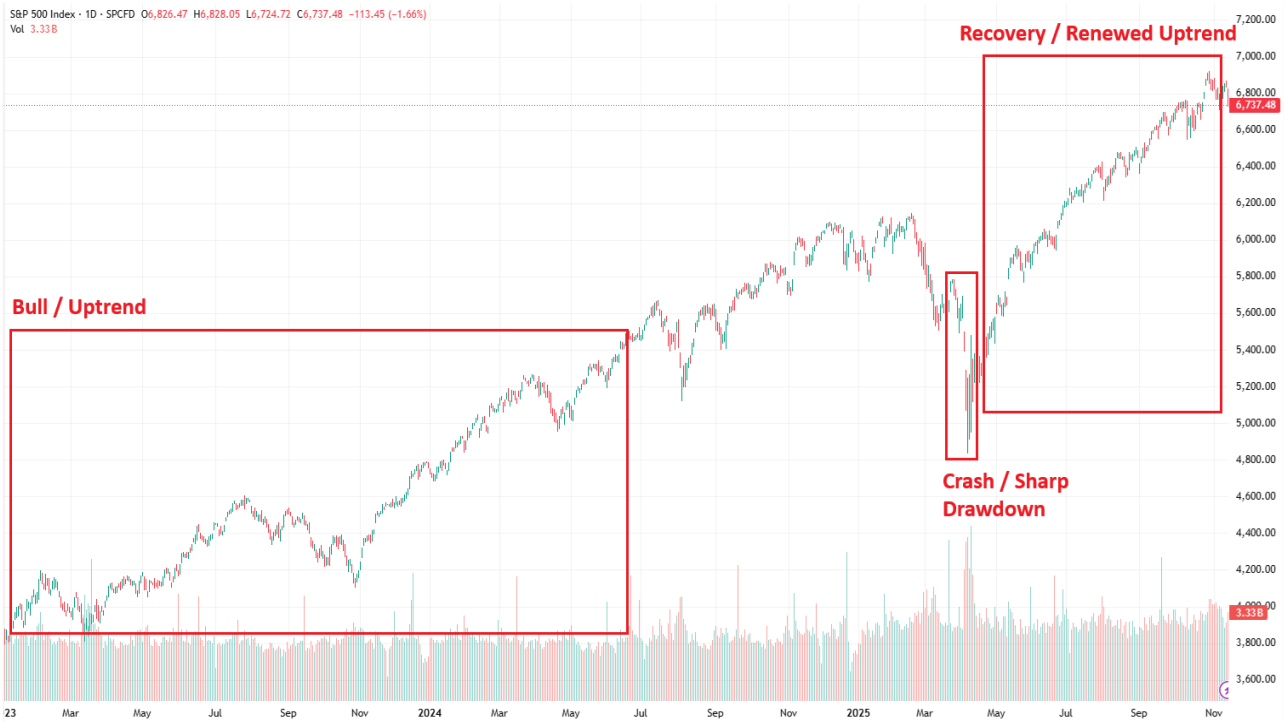

Figure. 4 Market Regime Timeline (2023-2025)

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The model adapted by dynamically adjusting positions, favoring cash or inverse exposure when volatility spiked, using multi-modal signals to detect early signs of market stress. MARAG-Fin achieved a prediction accuracy of 66.7%, a cumulative return of +7.43%, a Sharpe ratio of 2.21, and a maximum drawdown of −4.61%. Ablating the Chain-of-Thought (CoT) reasoning caused premature reversal predictions, increasing drawdowns, which highlights CoT's role in stabilizing decisions during turbulent periods. Overall, the model demonstrated an effective shift from maximizing predictive confidence to uncertainty-aware decision-making, crucial for mitigating risk in adverse market conditions.

6.4.3. Ablation on recovery market regime

In the recovery regime, which follows crashes and is characterized by gradual stabilization and re-emerging bullish momentum, MARAG-Fin showed strong adaptability in returning to opportunity-driven strategies. Recovery periods are typically driven by easing market fears, positive earnings surprises, economic stabilization, and renewed optimism in growth sectors. The model leveraged regime-detection signals and contextual memory from prior states to smoothly transition from defensive to opportunistic positioning. During this regime, MARAG-Fin achieved a prediction accuracy of 60%, a cumulative return of +5.59%, a Sharpe ratio of 1.14, and a maximum drawdown of −3.46%. Removing the ReAct contextual memory mechanism made the model less responsive to emerging momentum, showing that incorporating prior market context is important for capturing gains in transitional markets. Compared with baseline LLM agents lacking regime awareness, MARAG-Fin adjusted more quickly, highlighting the value of combining sentiment, context, and structured reasoning in post-crash environments.

6.4.4. Comparative discussion across market regimes

Across all three market regimes, MARAG-Fin consistently outperformed ablated versions of the model, maintaining higher Sharpe ratios and lower drawdowns, which demonstrates its ability to generalize across changing and non-stationary financial environments.

1. In the bull regime, the model effectively capitalized on sustained upward momentum driven by AI and tech optimism, favourable macroeconomic trends, and supportive monetary expectations, translating sentiment signals into long-biased, high-return strategies.
2. During the crash regime, MARAG-Fin adapted to heightened volatility and negative sentiment, shifting toward cash or inverse positions in response to sudden shocks such as trade or policy announcements. This risk-aware behaviour helped contain losses while maintaining stability.
3. In the recovery regime, the model leveraged contextual memory and regime detection to re-enter opportunistic positions, responding to stabilizing market signals, easing fears, and renewed bullish sentiment.

Overall, the combination of ReAct and CoT was key to strong performance across all market regimes. Without either component, the model lost accuracy, reacted more slowly, and traded less effectively. This shows that MARAG-Fin's reasoning-enhanced design helps it navigate everything from bull markets to crashes and recoveries.

6.5 Benchmarking MARAG-fin against state-of-the-art financial LLM agents

A comparative evaluation was conducted between MARAG-Fin and MarketSenseAI [30], a leading LLM-based financial agent that integrates textual and market data for short-term stock prediction. Both systems were tested on the S&P 500 index from January to December 2024 using identical conditions and metrics.

While MarketSenseAI shows a marginal advantage in risk-adjusted efficiency, MARAG-Fin demonstrates stronger overall performance characteristics that are more relevant for practical trading and portfolio management. Its ability to generate higher cumulative returns and maintain substantially lower drawdowns indicates more effective opportunity capture and superior downside protection. These outcomes suggest that MARAG-Fin's decision framework responds more adaptively to shifting market dynamics, allowing it to participate in favorable trends while avoiding deep losses during periods of heightened volatility. The model's consistency across market phases further implies a more robust internal reasoning structure, particularly in how it integrates multi-factor signals and adjusts exposure in real time.

Table 6. Performance Comparison of MARAG-Fin and MarketSenseAI on the S&P 500 (2024)

Framework	SR	CR	MDD
MARAG-Fin	2.17	30.27	5.77
MarketSenseAI	2.26	25.60	8.40

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The comparison highlights a key architectural difference between the agents. While MarketSenseAI focuses on incremental Sharpe Ratio improvements, MARAG-Fin balances growth with capital preservation, emphasizing stability. This makes it well-suited for institutional settings where large drawdowns can breach risk limits or operational constraints. Its lower volatility also supports risk-sensitive strategies such as long-only equities, factor-tilted portfolios, and systematic allocation. Overall, MARAG-Fin offers more resilient and consistent performance across market conditions, providing a reliable foundation for real-world decision-support.

7. Conclusion and future work

This study demonstrates that the proposed multi-agent Agentic RAG architecture provides an adaptive and accurate framework for document-based reasoning, stock price forecasting, and investment strategy formulation. By integrating multi-agent systems with Large Language Models (LLMs) and contextual retrieval mechanisms, the system enhances analytical precision and adaptability in complex financial environments. In terms of reasoning quality, ReAct Prompting achieved a RAGAS score of 0.92 (faithfulness = 0.86, relevance = 0.90, context precision = 1.00), significantly outperforming the Zero-shot approach. The GPT-4.1 model further improved performance with an aggregate score of 0.94, indicating that structured, stepwise reasoning enhances factual accuracy and contextual consistency in multi-agent decision-making. For stock price prediction, the NeuralProphet-powered dynamics agent produced stable forecasts with MAPE values of 0.038-0.039 for 1- and 5-day horizons and an RMSE of approximately 290, demonstrating reliability for short- and medium-term investment decisions. Backtesting results confirmed the practical efficacy of the framework: the S&P 500 strategy achieved a balanced return-risk profile (win rate 57.14%, total return 19.85%, Sharpe ratio 1.65, Sortino ratio 3.66, MDD 6.88%, CAGR 28.32%), whereas the NASDAQ-100 strategy yielded higher growth (total return 57.19%) at the expense of increased volatility (MDD 15.02%, drawdown duration 44 days).

These findings highlight that the effectiveness of a multi-agent system critically depends on high-quality LLM prompting and reasoning, ensuring context-aware, consistent, and faithful outputs. The integration of accurate predictive components further enhances the system's ability to support data-driven and adaptive investment decisions. Notably, strategies based on the S&P 500 demonstrate superior risk-adjusted performance, while NASDAQ-100 strategies cater to investors targeting higher growth under elevated risk conditions. Overall, the study confirms that intelligently integrated multi-agent architectures can deliver robust reasoning, reliable forecasting, and adaptive investment strategies, effectively balancing risk management with market opportunities.

Despite these promising results, several avenues for future research remain. Optimizing LLM retrieval and prompting strategies, particularly with emerging models such as GPT-5, is expected to further improve reasoning and decision quality. Future work will explore hybrid retrieval pipelines, LLM-based reranking, and adaptive prompting mechanisms. Expanding the forecasting horizon to incorporate macroeconomic, sentiment, and behavioural indicators, as well as extending to multi-asset portfolios including equities, bonds, commodities, and cryptocurrencies, can enhance predictive robustness and diversification. Additionally, real-time or paper-trading simulations, combined with advanced coordination strategies such as shared-reward mechanisms, may improve system adaptability and collective reasoning. Ensuring interpretability and transparency of agent outputs will remain critical for fostering trust and responsible deployment of Agentic RAG in financial decision support.

Data availability

The financial data used in this study were sourced from Yahoo Finance (https://finance.yahoo.com), which provides publicly accessible market information for research purposes. The processed datasets generated during the study are available in the authors' online repository at: https://github.com/skyradez/MARAG-Fin-Dataset.

Conflicts of Interest

The authors declare no financial or personal conflicts of interest related to this work.

Author Contributions

Conceptualization, Marvin Luckianto and Alexander Agung Santoso Gunawan; methodology, Marvin Luckianto and Alexander Agung Santoso Gunawan; validation, Marvin Luckianto and Alexander Agung Santoso Gunawan; formal analysis, Marvin Luckianto; investigation, Marvin Luckianto; resources, Marvin Luckianto and Alexander Agung Santoso Gunawan; data curation, Marvin Luckianto; writing-original draft preparation, Marvin Luckianto;

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writing-review and editing, Marvin Luckianto; visualization, Marvin Luckianto; supervision, Alexander Agung Santoso Gunawan; project administration, Alexander Agung Santoso Gunawan. All authors have read and agreed to the published version of the manuscript.

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